AN AGGREGATE CM-PAIR ESTIMATE AND A 5/3 BOUND FOR VALUATION PRODUCTS OF abc TRIPLES

RUIYI ZHANG

ABSTRACT. Let $a+b=c$ be pairwise coprime positive integers. Then, we introduce $R=\mathrm{rad}(abc)$ and $P(abc)=\prod_{p\mid abc} v_p(abc)$. We present here the proof of

$$P(abc),\ \mathbf{d}(abc) \ll_\varepsilon R^{5/3+\varepsilon},$$

where $\mathbf{d}$ is taken to be the divisor function. The first step in the deduction gives an aggregate fixed-radius estimate for CM points on an indefinite Shimura curve. In the case where the CM discriminants lie in $[\Delta, 2\Delta]$, the number of nearby ordered pairs is

$$\#\{(P,Q)\in \mathscr{P}_\Delta^2 : d(P,Q)\le r_0\} \ll |\mathscr{P}_\Delta| + \Delta\left(1+\frac{\Delta^2}{N}\right)N^\varepsilon,$$

which holds uniformly in all polynomial ranges $\Delta \le N^A$. This is established from the published local-density formulas, which are used with the characteristic-matrix framework of Bruinier, Li and Möller. The spatial Cauchy–Schwarz then yields a bound for the first absolute moment of the CM-packet. The ordinary AM–GM now applies directly to the logarithmic average in Pasten's Arakelov estimate to complete the argument, and neither a fourth moment nor a pointwise subconvexity estimate is required.

Date: July 2026.

2020 *Mathematics Subject Classification.* Primary 11G18; Secondary 11G05, 11D75, 11R29.

Key words and phrases. abc triples, Shimura curves, CM points, quaternion orders, valuations.

Independent Researcher, Columbia, Maryland, USA. Email: ryzfsu@gmail.com.

1. Introduction

Let a, b, c be pairwise coprime positive integers satisfying $a + b = c$. Here

$$R = \operatorname{rad}(abc), \qquad P(abc) = \prod_{p \mid abc} v_p(abc), \qquad \mathbf{d}(n) = \prod_{p \mid n} (v_p(n) + 1).$$

Using a comparison between classical and quaternionic modular degrees, Pasten was able to derive the unconditional bounds $P(abc), \mathbf{d}(abc) \ll_\varepsilon R^{8/3+\varepsilon}$ by evaluating an integral differential at CM points [12, Theorem 1.11 and Theorem 16.8]. Our main result is a reduction of the exponent.

Rickards has established precise formulas regarding the intersections of closed geodesics associated with pairs of real quadratic embeddings. These formulas include x-linking and local Eichler-order information [15, Theorems 10 and 63]. More recently, Bruinier–Li–Möller considered all signatures of a pair of quadratic vectors. More specifically, their formula gives the number of pairs of Heegner points at a given distance, and this number is a product of local factors [3, Theorem 1.1(iv) and Corollary 5.9]. We use Bruinier–Li–Möller's characteristic-matrix and distance formalism, but we derive our coarse orbit bound directly from the published results on the orbital integrals and local densities of Kudla–Rapoport–Yang [8, Proposition 5.3.6 and Chapter 5] and Yang's odd and dyadic formulas [17, Theorem 7.1] and [18, Theorem 4.4]. The aggregate estimate provides us with the divisor-sum bound we need for Pasten's argument.

In the context of the theory of special cycles, Heegner points, and quaternionic Shimura varieties, the natural objects in that framework are CM points [8, 21, 20, 19]. Complete or structured packets satisfy the equidistribution, variance, and small-scale bounds [10, 5, 7, 9, 1, 6, 16]. None of these works states our fixed-radius bound for arbitrary selected submultisets. Our approach is to substitute the classical sup-norm techniques [2, 4] with a local L^1 inequality. Later abc work treats different functionals or additional size hypotheses [13].

Theorem 1.1. *For every $\varepsilon > 0$ and for every pairwise coprime triple of positive integers $a + b = c$, we have*

$$P(abc),\ \mathbf{d}(abc) \ll_\varepsilon \operatorname{rad}(abc)^{5/3+\varepsilon}.$$

Geometrically speaking, let R_N stand for an Eichler order of reduced discriminant N in an indefinite quaternion division algebra, putting $X_N = R_N^1 \backslash \mathfrak{H}$. We assume that the odd part of N is square-free, and that $v_2(N) \leq 8$. For odd fundamental discriminants, $-d$ being coprime to N, and $d \in [\Delta, 2\Delta]$, choose $\mathscr{P}_\Delta$ as any multiset of the related CM centers where the multiplicity of each one does not exceed a fixed $B_0 \geq 1$.

Theorem 1.2 (Aggregate CM-pair estimate)**.** *Fix $A, r_0 > 0$ and $B_0 \geq 1$. If $\Delta \leq N^A$ and every multiplicity in $\mathscr{P}_\Delta$ is at most B_0, then*

$$\#\{(P, Q) \in \mathscr{P}_\Delta^2 : d(P, Q) \leq r_0\} \ll_{A, r_0, B_0, \varepsilon} |\mathscr{P}_\Delta| + \Delta \left(1 + \frac{\Delta^2}{N}\right) N^\varepsilon.$$

The estimate is uniform in the chosen discriminant submultiset.

For pairs lying on the off-diagonal, we will set $-d, -e$ to be the discriminants and b to be the integral pairing parameter. The local conditions state that the odd part of N has to divide $b^2 - de$. The published results on local densities allow us to compute an upper bound of N^{ε} for each orbit count. After we have fixed the radius we will be down to $O_{r_0}(\Delta(1+\Delta^2/N))$ choices for b and the quotient. These choices fix de, and de can have at most N^{ε} admissible factorizations. Further, the thinning of the packets considered here will only eliminate pairs.

In the case of $\mathscr{P}$, that is, the corresponding full Galois CM multiset from Pasten's construction, we use $x = N^{2/3+\delta}$ and $\vartheta = \delta/5$. Then, using theorem 1.2, the local holomorphic L^1 inequality, and the spatial form of the Cauchy–Schwarz inequality, we obtain

$$\frac{1}{H}\sum_{d}\frac{1}{2h_d}\sum_{P\in\mathscr{C}_d}|F_N(P)| \ll N^{\tau+\varepsilon}, \qquad \tau = \left(\frac{2}{3}+\delta\right)\frac{6\delta}{5}.$$

Here $\|F_N\|_{L^2(\mu_{\mathrm{prob}})} = N^{o(1)}$. Applying zero-counting and Markov's inequality at this point, we obtain a full packet that is free of zeros, on which the first absolute moment is $\ll N^{\tau+\varepsilon}$. We write $u_\sigma = |f(\tau_\sigma)|\,\mathrm{Im}\,\tau_\sigma > 0$, and for these we need only AM–GM:

$$\frac{1}{2h_d}\sum_{\sigma}\log u_\sigma \le \log\left(\frac{1}{2h_d}\sum_{\sigma}u_\sigma\right).$$

Using these first moment and AM–GM bounds along with Pasten's Proposition 14.2, height formula, and modular-degree comparison gives the cell exponent $q_\delta = 5/3+13\delta/5+12\delta^2/5$. In place of Pasten's norm exponent value of $5/6$, we can put the value $1/3 + o(1)$ that our aggregate first-moment estimate gives us. With that replacement, $1 + 2(5/6) = 8/3$ changes to $1 + 2(1/3) = 5/3$. Finally, the cyclic pairs and triples will distribute the level factors that are left, and we let $\delta \to 0$ to obtain the proof of theorem 1.1.

2. Quaternionic CM packets

Let B represent an indefinite quaternion division algebra over $\mathbb{Q}$, and let R_N denote an Eichler order of reduced discriminant $N = DM$, where D is the discriminant of B and M the Eichler level. We assume that the odd part of N is square-free and that $v_2(N) \le 8$. Write

$$\Gamma_N = R_N^1/\{\pm 1\}, \qquad X_N = \Gamma_N\backslash\mathfrak{H}.$$

In this notation we write $d\mu_{\mathrm{phys}} = dx\,dy/y^2$ for the hyperbolic measure in the physical sense, we let μ_{prob} be the probability measure, and we set $V_N = \mathrm{vol}(X_N)$. Using the exact volume formula from [3, Equation (2.7)], we get, for every $\varepsilon > 0$,

$$N^{1-\varepsilon} \ll_\varepsilon V_N \ll_\varepsilon N^{1+\varepsilon}.$$

As for the Gross lattice, we can write it out in the following form

$$L_N = \{2a - \mathrm{trd}(a) : a \in R_N\} \subset B^0.$$

In what follows we adopt the source convention $Q = -\,\mathrm{nrd}$ on the trace-zero space, with its full polarization

$$B_Q(x, y) = Q(x + y) - Q(x) - Q(y)$$

on the same space, so that $B_Q(x, y) = \mathrm{trd}(xy)$. We note that the primitive vectors representing imaginary quadratic discriminant $-d$ satisfy $Q(x) = -d < 0$. Taking lifts and signs for the associated negative lines, we read off the relation

$$\frac{|B_Q(x, y)|}{2\sqrt{Q(x)Q(y)}} = \cosh d(x, y).$$

In fact, if we divide x and y by the square roots of $-Q(x)$ and $-Q(y)$, we recognize the standard distance equation for the negative sheet of the hyperboloid model, the 2 being the contribution of the full polarization.

Lemma 2.1 (Lattice and characteristic-matrix normalization)**.** *The lattice L_N is exactly the index-four lattice L that we find in* [3, Equation (2.9)]*. An optimal embedding of the imaginary quadratic maximal order of odd fundamental discriminant $-d$ determines, up to sign, a primitive $x \in L_N$ with $Q(x) = -d$, and the converse holds as well. For a pair x, y, the characteristic matrix in the notation of Bruinier–Li–Möller is*

$$T = \frac{1}{2}\big(B_Q(x_i, x_j)\big)_{i,j} = \begin{pmatrix} -d & b \\ b & -e \end{pmatrix}, \qquad b = \frac{1}{2}B_Q(x, y).$$

Proof. If $a \in R_N$, we can check that either $2a - \mathrm{trd}(a)$ lies in $2(R_N \cap B^0)$ or $(1 + 2a - \mathrm{trd}(a))/2 \in R_N$, according to the parity of $\mathrm{trd}(a)$. In both cases the converse holds as well, and we obtain the equality with the lattice in [3, Equation (2.9)]. Let us assume now that $x \in L_N$ and $Q(x) = -d$, with $-d$ being an odd fundamental discriminant. Then x cannot belong to $2(R_N \cap B^0)$, therefore $\alpha = (1 + x)/2 \in R_N$. Since

$$\alpha^2 - \alpha + \frac{d + 1}{4} = 0,$$

α defines an embedding of the maximal order $\mathbb{Z}[(1 + \sqrt{-d})/2]$ into R_N, and this embedding is optimal since the source order is maximal. Conversely, an optimal embedding φ sends $\sqrt{-d}$ to the vector $x = \varphi(\sqrt{-d}) \in L_N$ with $Q(x) = x^2 = -d$. This vector is primitive. Indeed, $Q(L_N) \subset \mathbb{Z}$ because $Q(2a - \mathrm{trd}(a)) = \mathrm{trd}(a)^2 - 4\,\mathrm{nrd}(a)$ for $a \in R_N$, and hence a factorization $x = kx_0$ in L_N, with $k > 1$, would force $k^2 \mid d$, contrary to the square-freeness of d. This proves the stated correspondence, and it also proves the normalization in the characteristic matrix $\frac{1}{2}(B_Q(x_i, x_j))$. In addition, let us state the values $x = 2\alpha - 1$ and $y = 2\beta - 1$, where $\alpha, \beta \in R_N$ both have the reduced trace value of one. Then

$$B_Q(x, y) = 4\,\mathrm{trd}(\alpha\beta) - 2, \qquad b = 2\,\mathrm{trd}(\alpha\beta) - 1 \in \mathbb{Z},$$

so b is odd. This direct calculation fixes the normalization that we will use below, and we can check that its off-diagonal entry agrees with [3, Equation (2.14)]. □

In what follows, we will take CM discriminants to be coprime to N and to lie in $[\Delta, 2\Delta]$.

Remark 2.2. In the following, we will write h_d for the class number of $\mathbb{Q}(\sqrt{-d})$, while $\mathscr{C}_d$ will denote the complete $2h_d$-element Galois multiset of Pasten, and not a selected orientation packet. Moreover, we can use the fact that the CM point of Pasten has a residue field equal to the Hilbert class field [12, Section 13.1], which ensures that these $2h_d$ conjugates are distinct quotient points. Each of the elements is a CM center of discriminant $-d$, and hence we can see that it has the primitive-vector realization in lemma 2.1. Since the pair estimate is uniform for submultisets, we do not need to identify this Galois multiset with one prescribed system of local orientations. As for standard optimal-embedding counts with a fixed local class, we can find such counts stated in [11, Theorem 2.5].

Remark 2.3. For nonsingular T, we can recall that Bruinier–Li–Möller identify R_T with the diagonal norm-one orbit set of ordered pairs in L_N^2 that have characteristic matrix T. See their definition of R_T together with [3, Equation (2.9)]. The paragraph preceding their Theorem 5.7 explicitly states that the lattice in use for their subsequent local computations is reset to the localization of this same index-four lattice. Their count allows for the inclusion of imprimitive pairs while the CM centers are derived from primitive vectors, so we can see that the latter classes form a subset. The bound for the entire orbit set below relies on the published local-density formulas of Kudla–Rapoport–Yang, whereas we will not use the stronger, exact formula of Bruinier–Li–Möller as a logical input. Indeed, a CM center determines its negative rational line, on which a primitive vector of norm $-d$ is unique up to sign, since any two such vectors differ by a rational scalar with square equal to one. In moving to the quotient, at most an elliptic stabilizer can be added, and by the integral-trace argument in the proof of lemma 4.1, the projective order of that stabilizer is bounded by 3. Thus the fibers of the map from primitive-vector classes to CM centers are uniformly bounded, and the bounded multiplicities in $\mathscr{P}_\Delta$ will be absorbed in the implied constant.

3. The aggregate CM-pair estimate

We prove theorem 1.2.

Lemma 3.1 (Primitive local-factor bound)**.** *Let*

$$T = \begin{pmatrix} -d & b \\ b & -e \end{pmatrix}, \qquad N^\flat = N/2^{v_2(N)},$$

and we will assume that T is nonsingular. We let $R_T(D,M)$ denote the complete set of all norm-one equivalence classes with characteristic matrix T, for the order of discriminant D and level M. When the order is clear we write R_T for that set. If R_T is nonempty, then

$$N/N^\flat \le 2^8, \qquad N^\flat \mid b^2 - de. \tag{1}$$

Moreover, uniformly for $d, e \asymp \Delta$, $|b| \ll_{r_0} \Delta$, and $\Delta \le N^A$,

$$|R_T| \ll_{A,r_0,\varepsilon} N^\varepsilon. \tag{2}$$

Proof. Write $M = 2^k M_0$ with M_0 odd. For $k > 0$, let R_{DM_0} be the Eichler overorder of R_N of level M_0; admissibility gives $2 \nmid D$. Then

$$L_N \subset L_{DM_0}, \qquad \Gamma_N \subset \Gamma_{DM_0}.$$

Thus every Γ_{DM_0}-orbit splits into at most $[\Gamma_{DM_0} : \Gamma_N]$ many orbits of Γ_N. The exact volume formula therefore gives

$$|R_T(D,M)| \le [\Gamma_{DM_0} : \Gamma_N] |R_T(D,M_0)|, \qquad [\Gamma_{DM_0} : \Gamma_N] = 3 \cdot 2^{k-1} \le 384. \tag{3}$$

It remains to treat $M = M_0$ odd.

Use the standard identification $G = \mathrm{SO}(B^0) \simeq B^\times/\mathbb{Q}^\times$. Let $\mathcal{K}_p$ be the image of $R_{N,p}^\times$ in $G(\mathbb{Q}_p)$ and put $\mathcal{K} = \prod_p \mathcal{K}_p$. Let φ_p be the characteristic function of $L_{N,p}^2$ and $\varphi_f = \prod_p \varphi_p$. Fix a rational pair x_0 with Gram matrix T. Using the Witt extension, we see that every rational pair with Gram matrix T lies in its $G(\mathbb{Q})$-orbit. This is because an isometry between the two nondegenerate planes can be extended to the ambient space, and its determinant can then be corrected along the orthogonal line. The stabilizer of x_0 in G is trivial. The reason is that it fixes that plane pointwise and that it must act as $+1$ on the one-dimensional orthogonal complement of that plane.

In what follows we derive the precise coset form used here. With the Haar measures of [8, Section 5.3], triviality of the stabilizer gives

$$O_{T,p}(\varphi_p) = \int_{G(\mathbb{Q}_p)} \varphi_p(g_p^{-1} x_0)\, dg_p.$$

Since $\varphi_p(-z) = \varphi_p(z)$, the $\mathrm{O}(B^0)(\mathbb{Q}_p)$-orbital integral in the KRY comparison equals the $G(\mathbb{Q}_p)$-orbital integral used here. The integrand is right $\mathcal{K}_p$-invariant. If we decompose its support into right $\mathcal{K}_p$-cosets, each one of which has measure $\mathrm{vol}(\mathcal{K}_p)$, and taking the restricted product, this gives

$$\prod_p \frac{O_{T,p}(\varphi_p)}{\mathrm{vol}(\mathcal{K}_p)} = \#\{g\mathcal{K} \in G(\mathbb{A}_f)/\mathcal{K} : \varphi_f(g^{-1}x_0) = 1\}.$$

For almost every p, the data are unramified and $x_0 \in L_{N,p}^2$. The unramified maximal-lattice formula [8, Corollary 5.2.5], specialized below, gives $O_{T,p}(\varphi_p)/\mathrm{vol}(\mathcal{K}_p) = 1$. The support is a union of right $\mathcal{K}_p$-cosets, each of them contributing one after division by $\mathrm{vol}(\mathcal{K}_p)$. Note also that $\mathcal{K}_p$ itself lies in the support. Then we conclude that the support is precisely that

single coset. Thus both the product and the adelic support above are finite restricted products. In the nonsingular case, we see that the local supports contain finitely many such cosets. The corresponding component-sum identity we quote directly from [8, Proposition 5.3.6].

By full-unit rational orbit set we mean equivalence under the image of $R_N^\times$ in $G(\mathbb{Q})$. As we check below, it maps into the adelic set on the right. Then we see that this map sends the equivalence class of $x = g^{-1}x_0$, with $g \in G(\mathbb{Q})$, to $g\mathcal{K}$. This map is injective and well-defined because $G(\mathbb{Q}) \cap \mathcal{K}$ is the image of $R_N^\times$. In order to demonstrate the latter assertion, we may take a class in $G(\mathbb{Q}) \cap \mathcal{K}$ and represent it as $q \in B^\times(\mathbb{Q})$. For each p write $q = \lambda_p u_p$, with $\lambda_p \in \mathbb{Q}_p^\times$ and $u_p \in R_{N,p}^\times$, and put $n_p = v_p(\lambda_p)$. Taking reduced norms gives $v_p(\mathrm{nrd}(q)) = 2n_p$ because $\mathrm{nrd}(u_p) \in \mathbb{Z}_p^\times$, so we see that all but finitely many of the n_p vanish. For $s = \prod_p p^{-n_p} \in \mathbb{Q}^\times$, we have $s\lambda_p \in \mathbb{Z}_p^\times$, which gives $sq \in R_{N,p}^\times$ for all p. The same holds for $(sq)^{-1}$. Since $R_N = B(\mathbb{Q}) \cap \prod_p R_{N,p}$, we conclude that $sq, (sq)^{-1} \in R_N$, and hence that $sq \in R_N^\times$. Finally, we note that the norm-one image lies in the full-unit image with index at most two. Hence,

$$|R_T(D, M_0)| \le 2 \prod_p \lambda_p(T), \qquad \lambda_p(T) = \frac{O_{T,p}(\varphi_p)}{\mathrm{vol}(\mathcal{K}_p)}, \tag{4}$$

where $O_{T,p}$ denotes the local orbital integral calculated with the Haar measures of [8, Section 5.3]. At all the unramified primes discussed previously, $\lambda_p(T) = 1$. In this argument, we need no assertions about the connected component or the type-number.

We now record how the published local formulas specialize in the case at hand. Put $t = v_p(\det T)$. At an odd prime let p^{a_1}, p^{a_2} be the elementary divisors of T. Square-freeness of d, e gives $a_1 \le 1$, and $(de, N) = 1$ gives $a_1 = 0$ at $p \mid N$. Using the orbital integral/Whittaker comparison together with the maximal-lattice formula of [8, Proposition 5.3.3, Lemma 5.3.9, and Corollary 5.2.5], we read off, for the odd primes $p \nmid DM_0$, the value

$$\lambda_p(T) \le t + 1 \quad (a_1 = 0), \qquad \lambda_p(T) = 2 \quad (a_1 = 1),$$

whenever T is represented, and zero otherwise. In the second case $t \ge 2$. When p is odd and $p \mid D$, the ramified formula [8, Theorem 5.7.2 and Lemmas 5.3.9 and 5.7.1] shows that $\lambda_p(T) = 2$ whenever it is nonzero, and nonvanishing forces $t \ge 1$.

If $p \mid M_0$, then $v_p(M_0) = 1$. The local trace-zero Eichler lattice is then the lattice carrying the quadratic form $a^2 + pbc$. Since $a_1 = 0$, write $T \sim \mathrm{diag}(\varepsilon_1, \varepsilon_2 p^t)$. Substituting $\langle 1, p, -p\rangle$ and this T into [17, Theorem 7.1] gives

$$\alpha_p(T, L_{N,p}) = (p-1)t(1 + (\varepsilon_1, p)_p).$$

Here $[L_{N,p}^\# : L_{N,p}] = p^2$ and $\mathcal{K}_p$ has index $p+1$ in the maximal compact. Since the Haar normalization of KRY is in force here, we can record that $\mathrm{vol}(\mathcal{K}_p^{\max}) = 1 - p^{-2}$ and hence that $\mathrm{vol}(\mathcal{K}_p) = (p-1)/p^2$, while Lemma

5.7.1 gives us $O_{T,p} = \alpha_p/p^2$. Therefore $\lambda_p(T) = 2t$ or 0. In particular, nonvanishing forces $t \geq 1$, and

$$0 \leq \lambda_p(T) \leq 2t \leq (t+1)^2. \tag{5}$$

At $p = 2$, the odd fundamental-discriminant condition and lemma 2.1 between them will give us $d \equiv e \equiv 3 \pmod 4$, b odd, and $t = v_2(\det T) \geq 2$. Splitting off the unit entry $-d$ gives

$$T \sim \operatorname{diag}(\epsilon_1, 2^t\epsilon_2), \qquad \epsilon_i \in \mathbb{Z}_2^\times, \qquad \epsilon_1 \equiv 1 \text{ or } 5 \pmod 8.$$

Since the level is odd, the two local Gross forms, along with their dual indices and the KRY compact volumes, are

	$[L_{N,2}^{\#} : L_{N,2}]$	$\operatorname{vol}(\mathcal{K}_2)$
$z^2 + 4uv \quad (2 \nmid D)$	2^5	$3/8$
$-3z^2 - 8(x^2 + xy + y^2) \quad (2 \mid D)$	2^7	$3/2$.

Here the volumes are [8, Lemma 5.3.9]. Put

$$c_t(\epsilon_1) = \begin{cases} \max(t-1, 0), & \epsilon_1 \equiv 1 \pmod 8, \\ 1, & \epsilon_1 \equiv 5 \pmod 8. \end{cases}$$

When we substitute the two Gross forms shown above and $(a_1, a_2) = (0, t)$ into the dyadic formula [18, Theorem 4.4] from Yang's work, the result is

$$\alpha_2(T, L_{N,2}) = \begin{cases} 12c_t(\epsilon_1), & 2 \nmid D, \\ 96c_t(\epsilon_1), & 2 \mid D, \end{cases} \tag{6}$$

whenever the Hasse invariants of both T and the ambient ternary space are equal, and zero otherwise. As for the normalized Whittaker factor in [3, Theorem 5.11], we can read from there that it is given by $\beta_2(T) = c_t(\epsilon_1)$. In the KRY orbital normalization used here, $\lambda_2(T) = \beta_2(T)$ in the split case, and $\lambda_2(T) = 2\beta_2(T)$ in the ramified case. As we can see, this comes from the conversion recorded below.

For completeness, we now record the normalization to the factor in (4). The local Siegel–Weil comparison [8, Proposition 5.3.3, Equation (5.3.46), and Lemma 5.7.1] gives

$$O_{T,2} = \frac{|D|_2^{-2}}{[L_{N,2}^{\#} : L_{N,2}]} \alpha_2(T, L_{N,2}).$$

As we can check, the $|2|_2^{1/2}$ factors drop out of both the density–Whittaker and the Whittaker–orbital comparisons. In the first row this gives $O_{T,2} = 12c_t/2^5 = (3/8)c_t$; in the second it gives $O_{T,2} = 4 \cdot 96c_t/2^7 = 3c_t$, since $|D|_2 = 1/2$. Finally, we see that dividing by the compact volumes gives

$$\lambda_2(T) = \begin{cases} c_t(\epsilon_1), & 2 \nmid D, \\ 2c_t(\epsilon_1), & 2 \mid D, \end{cases}$$

in all represented cases and zero elsewhere. In particular, we conclude that $\lambda_2(T) \leq 2(t+1) \ll (t+1)^2$ carries an implicit absolute constant.

In addition to this, we can also note that the same representation conditions imply that any odd prime factor which divides DM_0 must necessarily divide $\det T = de - b^2$ as well. We can now combine the preceding bounds in (4), prime by prime. In that way we can write

$$|R_T(D, M_0)| \ll \prod_{p|\det T} \left(v_p(\det T) + 1\right)^2 = \mathbf{d}(|\det T|)^2.$$

Given that (3) holds, and also since both $|\det T| \ll_{r_0} \Delta^2$ and $\Delta \leq N^A$ hold, we can then establish (2) from the standard divisor bound. □

Proof of theorem 1.2. For each ordered pair of quotient centers, we will need to choose lifts that realize the respective quotient distance after shifting one of the two by a member of Γ_N. In turn, we see that the corresponding simultaneous Γ_N-class of primitive vectors is counted by the appropriate R_T. By lemma 2.1, we see that its characteristic matrix is

$$T = \begin{pmatrix} -d & b \\ b & -e \end{pmatrix}.$$

We need to select signs for the two primitive vectors independently in such a way that $b > 0$. Then, from the hyperboloid calculation preceding lemma 2.1, we can read off $\cosh d(P, Q) = b/\sqrt{de}$. Also, we can switch either sign, replacing b by $-b$, but the CM center remains unchanged. Then we will find that the sign-invariant form is

$$\cosh d(P, Q) = \frac{|b|}{\sqrt{de}}. \tag{7}$$

We let R_T denote the complete set of norm-one equivalence classes with characteristic matrix T, and we write $N^\flat = N/2^{v_2(N)}$. We will treat the singular and the nonsingular cases separately.

If T is singular, then we can see that (7) gives distance zero. Since d, e are both fundamental, we see that $b^2 = de$ also gives $d = e$. Hence we can conclude that the corresponding quotient centers agree. With $m(P)$ the multiplicity of a center, we can read off from the bounded multiplicity that $\sum_P m(P)^2 \leq B_0 \sum_P m(P) = B_0 |\mathscr{P}_\Delta|$. Together with the bounded fibres in remark 2.3, we see that this bound provides a contribution given by $O_{B_0}(|\mathscr{P}_\Delta|)$. For a nonsingular pair we will need lemma 3.1. If the pair is at distance at most r_0, then we see that (7) gives $O_{r_0}(\Delta)$ possibilities for b. We can now put

$$m = \frac{b^2 - de}{N^\flat}.$$

Then, from Equations (7) and (1), we can read off

$$1 \leq m \ll_{r_0} 1 + \frac{\Delta^2}{N}.$$

For fixed b and m, we see that the product $de = b^2 - N^\flat m$ is fixed as well. Then we can bound the number of ordered factor pairs $d, e \in [\Delta, 2\Delta]$ by

its divisor function value. That divisor function value is N^{ε} throughout the polynomial range. The conditions on fundamental discriminants and coprimality thin this set of pairs, but as we can see they add nothing. The summation of (2) over b and m therefore leads us to

$$\ll_{A,r_0,\varepsilon} \Delta \left(1 + \frac{\Delta^2}{N}\right) N^{\varepsilon}$$

off the diagonal. Restricting to submultisets removes pairs and does nothing else, apart from the bounded multiplicities that were already controlled. This proves the theorem. □

4. A first absolute CM-packet moment

We now place theorem 1.2 in Pasten's family. For an abc triple, let N be the conductor of the Frey curve attached to that triple, and choose an admissible factorization $N = DM$. Let $0 < \delta < 1/10$ be a parameter to be fixed at the end. Pasten's sieve [12, Theorem 11.1 and Corollary 11.2], along with

$$x = N^{2/3+\delta}, \qquad \vartheta = \delta/5, \tag{8}$$

produces H odd fundamental discriminants $-d$, with $x < d \le 2x$, in which all the primes dividing D are inert while all the primes dividing M are split in $\mathbb{Q}(\sqrt{-d})$, and

$$H \gg x^{1-\delta}, \qquad d^{1/2-\vartheta} < h_d \ll_{\varepsilon} x^{1/2+\varepsilon}, \qquad \frac{L'}{L}(1,\chi_d) \ll \log\log d. \tag{9}$$

In particular, all these discriminants are coprime to N. In this situation we can apply the corollary that we cited, because $x = N^{2/3+\delta} > N^{1/2+\delta}$, and we can note that its constants are uniform in the admissible factorization $N = DM$. Let $\mathscr{C}_d$ be the full $2h_d$-element Galois multiset in remark 2.2, and set

$$\mathscr{P} = \bigsqcup_d \mathscr{C}_d, \qquad P = |\mathscr{P}|.$$

Then $P \ll Hx^{1/2+\varepsilon}$. Put $C(\mathscr{P}) := \#\{(P,Q) \in \mathscr{P}^2 : d(P,Q) \le 2r_1\}$, where r_1 is the fixed radius in lemma 4.1. Theorem 1.2 gives

$$C(\mathscr{P}) \ll N^{\varepsilon}\left(P + x + \frac{x^3}{N}\right). \tag{10}$$

Lemma 4.1 (Uniform local L^1 evaluation)**.** *Let $f(z)\,dz$ be a holomorphic differential on X_N, and we put*

$$F_N(z) = \frac{\sqrt{N}}{\|f\|_{\mathrm{Pas}}}\, y f(z), \qquad \|f\|_{\mathrm{Pas}}^2 = \int_{X_N} |y f(z)|^2 \, d\mu_{\mathrm{phys}}(z).$$

We regard F_N as a unitary automorphic section. Its pointwise norm $|F_N|$ is a well defined function on X_N, and we note that every evaluation and every

integral below involving F_N refers either to this norm or to its powers. For a hyperbolic radius r_1 that is fixed and sufficiently small, we have

$$|F_N(P)| \ll V_N \int_{B(P,r_1)} |F_N(z)|\, d\mu_{\mathrm{prob}}(z),$$

uniformly for the levels in section 2. Moreover we have $\|F_N\|^2_{L^2(\mu_{\mathrm{prob}})} = N/V_N$.

Proof. Fix a lift $\widetilde{P} = x_0 + iy_0$. Using the holomorphic submean inequality on a Euclidean disk of radius ρy_0, we read off, on a comparable hyperbolic disk,

$$|y_0 f(\widetilde{P})| \ll \int_{B(\widetilde{P},r_1)} |y f(z)|\, d\mu_{\mathrm{phys}}(z),$$

since $y \asymp y_0$ and $d\mu_{\mathrm{phys}} = dx\, dy/y^2$ hold there.

Next we need to make the descent to the orbifold uniform. The groups Γ_N are cocompact, so we conclude that they contain no parabolic elements. Also, a nonelliptic element has an integral reduced trace whose absolute value is at least 3, and so we conclude that its translation length is at least $\ell_0 = 2\,\mathrm{arcosh}(3/2)$. We let μ_2 denote a Margulis constant for the hyperbolic plane, and choose $r_1 < \min\{\mu_2/2, \ell_0/4\}$. Applying the two-dimensional Margulis lemma [14] and then the classification of elementary Fuchsian groups, we conclude that the subgroup generated by

$$\{\gamma \in \Gamma_N : d(\widetilde{P}, \gamma\widetilde{P}) < 2r_1\}$$

is finite. In fact, we note that an infinite elementary subgroup would either contain a parabolic element or preserve a geodesic. In the latter case, for the subgroup generated by the displayed set to have infinitely many elements, we need either a hyperbolic generator or two distinct half-turn generators. As for the hyperbolic generator, it would contradict the length gap, whereas we can also observe here that the product of the two distinct half-turn generators has a translation length shorter than $4r_1 < \ell_0$. Thus the small-displacement subgroup is finite. A finite subgroup of $\mathrm{PSL}_2(\mathbb{R})$ is cyclic. Also, any nontrivial torsion lift in this situation has its integral reduced trace lying in $\{-1, 0, 1\}$, so its projective order is no greater than 3. As a consequence, we can say that the projection of $B(\widetilde{P}, r_1)$ to X_N has a uniformly bounded multiplicity, because an overlap of $B(\widetilde{P}, r_1)$ with its γ-translate forces the inequality $d(\widetilde{P}, \gamma\widetilde{P}) < 2r_1$. Thus, when we descend the submean inequality that precedes this one, we can see that the effect is that of changing just an absolute constant. Finally, we can take $d\mu_{\mathrm{phys}} = V_N d\mu_{\mathrm{prob}}$ together with the definition of F_N, and in this way we read off the estimate that was asserted. The L^2 identity is immediate from the normalizations. □

Define

$$T_d(F_N) = \frac{1}{2h_d} \sum_{P \in \mathscr{C}_d} |F_N(P)|$$

and the sampling multiplicity $\mu(z) = \sum_{P \in \mathscr{P}} \mathbf{1}_{B(P,r_1)}(z)$. Since any two intersecting balls have their centers at a bounded distance from one another, we can see that

$$\int \mu^2 \, d\mu_{\mathrm{prob}} \ll \frac{C(\mathscr{P})}{V_N}.$$

Indeed, after we have expanded the square, we will find that the pairs of centers that contribute are just those whose distance apart is at most $2r_1$, and we can also see that each ball intersection has a probability measure that is equal to $O(1/V_N)$. If we now sum lemma 4.1, and then make use of the lower bound for h_d, and after that apply Cauchy–Schwarz in space, what we read off is the exact volume cancellation, and it comes out in the form

$$\begin{aligned} \frac{1}{H} \sum_d T_d(F_N) &\ll \frac{V_N}{H x^{1/2-\vartheta}} \int \mu |F_N| \, d\mu_{\mathrm{prob}} \\ &\ll \frac{\sqrt{NC(\mathscr{P})}}{H x^{1/2-\vartheta}}. \end{aligned} \tag{11}$$

If we now insert (10) together with the bound $P \ll H x^{1/2+\varepsilon}$, what we are left with is three terms:

$$E_{\mathrm{diag}} = N^{1/2} H^{-1/2} x^{-1/4+\vartheta+\varepsilon}, \quad E_x = N^{1/2} H^{-1} x^{\vartheta+\varepsilon}, \quad E_{\mathrm{off}} = H^{-1} x^{1+\vartheta+\varepsilon}.$$

Under (8)–(9), we can compute that the N-exponents of these three are

$$-\frac{17}{60}\delta + \frac{7}{10}\delta^2, \qquad -\frac{1}{6} - \frac{1}{5}\delta + \frac{6}{5}\delta^2, \qquad \tau = \left(\frac{2}{3} + \delta\right)\frac{6\delta}{5}.$$

It is the last of these three terms that is the limiting one here, and in this way we obtain the following proposition.

Proposition 4.2. *In the notation that we have set up in the preceding discussion,*

$$\frac{1}{H} \sum_d T_d(F_N) \ll N^{\tau+\varepsilon}, \qquad \tau = \left(\frac{2}{3} + \delta\right)\frac{6\delta}{5}.$$

5. The arithmetic conclusion

First we will need to choose a suitable packet on which logarithms can be taken. Using [12, Lemma 14.3], we can see that if the corresponding rational differential vanishes at one of the CM conjugate points, then it vanishes on the entire Galois orbit as well. Also, we can note that, by the same lemma, the associated modular form has at most $N^{1+\eta}$ zeros on $\mathfrak{H}$, under Γ_N-equivalence, for each fixed $\eta > 0$. Thus, for $\eta = \delta/5$, we can estimate the number B of bad discriminants as

$$B \ll N^{1+\delta/5} x^{-1/2+\delta/5}, \qquad \frac{B}{H} \ll N^{-\delta/2+6\delta^2/5} = o(1).$$

Since the negative line of a CM center determines the stabilizing torus of that CM center, which also determines the imaginary quadratic field, we see that all of the CM orbits corresponding to different quadratic fields are pairwise disjoint. Using Markov's inequality together with the estimate from proposition 4.2, we can get a positive proportion of quadratic fields with $T_d(F_N) \ll N^{\tau+\varepsilon}$. Hence we can be sure that one of them is also zero-free. We now fix such a field K. In fact, this field belongs to the original sieve, and we can also note that it keeps the logarithmic-derivative bound that was recorded in (9).

In what follows we will let $f(z)\,dz = f_{D,M}(z)\,dz$ be the integral quaternionic differential in Pasten's construction, that is to say, the differential which comes from the modular parametrization in the form in which he records it in his Equation (16.3). For all of the conjugates P_σ, and with a lift $\tau_\sigma \in \mathfrak{H}$ chosen for each one of them, we will write $u_\sigma = |f(\tau_\sigma)| \operatorname{Im} \tau_\sigma > 0$. Then, from the normalization that was recorded in lemma 4.1, we can read off $u_\sigma = \|f\|_{\mathrm{Pas}} |F_N(P_\sigma)|/\sqrt{N}$. Now, using ordinary AM–GM together with the first-moment estimate that was selected above, we can then go on to obtain

$$\frac{1}{2h_K} \sum_\sigma \log u_\sigma \leq \log \|f\|_{\mathrm{Pas}} - \frac{1}{2} \log N + (\tau + \varepsilon) \log N + O(1). \tag{12}$$

Also, Proposition 14.2 in Pasten is given with respect to exactly this hyperbolic metric, and therefore we can see that it implies

$$h_{\mathrm{Ar}}(P_K) \geq -\log(2M) - \frac{1}{2h_K} \sum_\sigma \log u_\sigma.$$

Now, using his height formula [12, Theorem 13.2] in the form in which it is recorded there, after the functional equation has been applied, we can go on to obtain

$$h_{\mathrm{Ar}}(P_K) = \frac{L'}{L}(1, \chi_K) + \frac{1}{2} \log(dD) + O(1).$$

Then, combining these formulas, and using $N = DM$ together with $d \leq 2N^{2/3+\delta}$, and together with the logarithmic-derivative estimate that was

retained above, we can now go on to deduce that

$$-\log\|f\|_{\mathrm{Pas}} \leq \frac{1}{2}\log M + \left(\frac{1}{3}+\frac{\delta}{2}+\tau+\varepsilon\right)\log N. \tag{13}$$

In preparation for the final step of the argument, we will now single out the one consequence of Pasten's modular-degree comparison that is used there.

Proposition 5.1 (Pasten interface)**.** *For the Frey branch and for an admissible factorization $N = DM$ which has the property that M is divisible by at least two odd primes, we have the following estimate*

$$\prod_{p\mid D} v_p(\Delta_E) \ll_\varepsilon N^{1+\varepsilon}\|f\|_{\mathrm{Pas}}^{-2}.$$

Proof. Now, from Pasten's Theorem 6.1(b.2), what we can read off is

$$\log\prod_{p\mid D} v_p(\Delta_E) \leq \log\delta_{1,N} - \log\delta_{D,M} + O(\omega(D)).$$

Using the degree relations (16.1)–(16.3), which are combined in [12, Theorem 16.1, pp. 305–307] and invoked in his Theorem 16.5(ii), we can then go on to read off the estimate

$$\log\prod_{p\mid D} v_p(\Delta_E) \leq 2\log|c_f| + 2\log\|f_{1,N}\|_2 - 2\log\|f_{D,M}\|_2 + O(\log\log N + \omega(D)).$$

Now the minimal isogeny has degree at most 163 by an application of the Mazur–Kenku theorem, in the form in which it is recorded in [12, proof of Theorem 16.1]. Thus the difference between the two terms which are associated with the Faltings height is $O(1)$. In the same normalization, the classical estimate of $\|f_{1,N}\|_2^2 \ll N\log N$, the Manin constant estimate that is due to Pasten [12, Theorem 1.1 and proof of Theorem 16.1], and the bound $\omega(D) \ll \log N/\log\log N$ together imply that for sufficiently large N the right-hand side is at most

$$(1+\varepsilon)\log N - 2\log\|f_{D,M}\|_2.$$

Finally, the non-normalized Petersson norms of Pasten take into account the same physical hyperbolic measure [12, Section 4.6], and therefore we can conclude that $\|f_{D,M}\|_2 = \|f\|_{\mathrm{Pas}}$. This proves the statement while additionally showing explicitly that the Frey branch contributes no further factor of D. □

For every factorization satisfying the hypothesis of proposition 5.1, applying it to (13) gives us the cell estimate

$$\prod_{p\mid D} v_p(\Delta_E) \ll_\varepsilon N^{q_\delta+\varepsilon}M, \qquad q_\delta = 1+\left(\frac{2}{3}+\delta\right)+2\tau = \frac{5}{3}+\frac{13}{5}\delta+\frac{12}{5}\delta^2. \tag{14}$$

It remains to distribute the factor M. In what follows we write the odd part of N as $p_1 \cdots p_m$, but the bounded factor 2^r remains. If m exceeds a fixed cutoff, then we can be sure that all subsequent levels contain at least two odd primes, as required in proposition 5.1. In the even case $m-2$ is even, and in the odd case $m-3$ is even. As a result, each D_j described below is an admissible indefinite quaternion discriminant and after the fixed cutoff has been increased, will exceed 1. Moreover $(D_j, M_j) = 1$, and all constants are uniform in j. We then take each entry (D_j, M_j) and apply to it the sieve, the pair estimates, the first-moment selection, and the Pasten interface. Note that the selected field can depend on j and that the earlier established uniformity makes the cell estimate independent of j. If m is even, apply (14) cyclically with

$$M_j = 2^r p_j p_{j+1}, \qquad D_j = \frac{p_1 \cdots p_m}{p_j p_{j+1}}.$$

We can note that all valuations occur on $m-2$ left-hand sides, and that all odd primes are included in two M_j. If $Q_{\mathrm{odd}} = \prod_{i=1}^{m} (2 v_{p_i}(abc)) = \prod_{i=1}^{m} v_{p_i}(\Delta_E)$, multiplication gives

$$Q_{\mathrm{odd}} \ll N^{q_\delta + (2q_\delta+2)/(m-2) + O(\varepsilon)}. \tag{15}$$

If m is odd, use cyclic triples

$$M_j = 2^r p_j p_{j+1} p_{j+2}, \qquad D_j = \frac{p_1 \cdots p_m}{p_j p_{j+1} p_{j+2}},$$

which gives

$$Q_{\mathrm{odd}} \ll N^{q_\delta + (3q_\delta+3)/(m-3) + O(\varepsilon)}. \tag{16}$$

If $m \le m_0$ for the chosen fixed cutoff, then we can see that $\omega(abc) = m+1 \le m_0+1$. We can first employ Pasten's Proposition 15.1 with a smaller internal epsilon and then select the maximum over these finitely many values, which gives $\mathbf{d}(abc) \ll_{\varepsilon, m_0} R^{1+\varepsilon}$ [12, Proposition 15.1]. From his Lemma 15.2 we can read off $v_2(abc) \ll_\varepsilon R^\varepsilon$ [12, Lemma 15.2 and Equation (16.4)], and the Frey conductor satisfies $N \asymp R$. Finally,

$$P(abc) \le \mathbf{d}(abc) \le (v_2(abc)+1) Q_{\mathrm{odd}}.$$

In this last step we needed to use the inequality $v+1 \le 2v$, and we needed it for each positive integer value v at the odd primes in question.

Given $\varepsilon_0 > 0$, we first need to choose a fixed $0 < \delta < 1/10$ for which $13\delta/5 + 12\delta^2/5 < \varepsilon_0/4$ holds. In the next step we need to ensure that the combined internal epsilon loss is below $\varepsilon_0/4$ and we need to select the prime-count cutoff such that both the overheads in (15)–(16) fall below $\varepsilon_0/4$ above it. Here we can rely on Proposition 15.1 of Pasten, which takes care of those cases where m is equal to or lower than our cutoff. The finitely many conductors below the previously established thresholds end up in the implied constant, and we do not track the effective dependence of that constant on ε_0. With that we have proved theorem 1.1.

Remark 5.2. We attain the exponent 5/3 in the proof by means of the discriminant scale $\Delta = N^{2/3+o(1)}$. We see that the above reasoning does not prove the abc conjecture. In fact, we note that it handles only the product of valuation exponents and the divisor function, not *abc* itself.

Notation

$R = \mathrm{rad}(abc)$	radical of the *abc* triple
$P(abc)$, $\mathbf{d}(n)$	valuation product and divisor function
B	indefinite quaternion division algebra over $\mathbb{Q}$
B (in §5)	number of bad discriminants
B^0, $Q = -\mathrm{nrd}$	trace-zero space and its quadratic form
B_Q	full polarization of Q, so $B_Q(x,y) = \mathrm{trd}(xy)$
D	discriminant of B
M	Eichler level
$N = DM$	reduced discriminant of R_N, and the Frey conductor
R_N	Eichler order of reduced discriminant N
$\Gamma_N = R_N^1/\{\pm 1\}$	norm-one units modulo ± 1
$\mathfrak{H}$	upper half plane
$X_N = \Gamma_N \backslash \mathfrak{H}$	the Shimura curve
$V_N = \mathrm{vol}(X_N)$	its hyperbolic volume
$L_N \subset B^0$	Gross lattice of [3, Equation (2.9)]
$G = \mathrm{SO}(B^0)$	$\simeq B^\times/\mathbb{Q}^\times$
$\mathcal{K}_p$, $\mathcal{K} = \prod_p \mathcal{K}_p$	image of $R_{N,p}^\times$ in $G(\mathbb{Q}_p)$, and its product
T, $R_T(D,M)$	characteristic matrix, and its norm-one class set
$\lambda_p(T)$, α_p	local orbital factor and local density
$-d$, $-e$	odd fundamental CM discriminants
h_d	class number of $\mathbb{Q}(\sqrt{-d})$
$\mathscr{C}_d$	the complete $2h_d$-element Galois multiset
Δ	the CM discriminant scale
$\mathscr{P}_\Delta$	a multiset of CM centers with $d \in [\Delta, 2\Delta]$
B_0	bound on multiplicities in $\mathscr{P}_\Delta$
$\mathscr{P} = \bigsqcup_d \mathscr{C}_d$	the full Galois CM multiset
$C(\mathscr{P})$	number of ordered pairs of $\mathscr{P}$ within $2r_1$
H	number of discriminants produced by Pasten's sieve
$x = N^{2/3+\delta}$, $\vartheta = \delta/5$	sieve parameters
$f(z)\,dz$	holomorphic differential on X_N, $f = f_{D,M}$ in §5
$\|f\|_{\mathrm{Pas}}$	its non-normalized Petersson norm
F_N	the unitary normalization $\sqrt{N}\, yf/\|f\|_{\mathrm{Pas}}$
$T_d(F_N)$	first absolute moment of $\lvert F_N\rvert$ over $\mathscr{C}_d$
$\mu(z)$	sampling multiplicity of the balls $B(P, r_1)$
r_0, r_1	fixed hyperbolic radii
τ, q_δ	moment exponent and cell exponent
Δ_E	discriminant of the Frey curve E

References

1. Montserrat Alsina and Dimitrios Chatzakos, *CM-points and lattice counting on arithmetic compact Riemann surfaces*, Journal of Number Theory **212** (2020), 339–353.
2. Valentin Blomer and Roman Holowinsky, *Bounding sup-norms of cusp forms of large level*, Inventiones Mathematicae **179** (2010), no. 3, 645–681.
3. Jan Hendrik Bruinier, Yingkun Li, and Martin Möller, *Intersections of special cycles on Shimura curves and Siegel Maass forms*, 2026, arXiv:2602.16248v1 [math.NT], February 2026.
4. Soumya Das and Jyoti Sengupta, *L^∞ norms of holomorphic modular forms in the case of compact quotient*, Forum Mathematicum **27** (2015), no. 4, 1987–2001.
5. Gergely Harcos and Philippe Michel, *The subconvexity problem for Rankin–Selberg L-functions and equidistribution of Heegner points. II*, Inventiones Mathematicae **163** (2006), 581–655.
6. Peter Humphries and Maksym Radziwiłł, *Optimal small scale equidistribution of lattice points on the sphere, Heegner points, and closed geodesics*, Communications on Pure and Applied Mathematics **75** (2022), 1936–1996.
7. Ilya Khayutin, *Joint equidistribution of CM points*, Annals of Mathematics **189** (2019), no. 1, 145–276.
8. Stephen S. Kudla, Michael Rapoport, and Tonghai Yang, *Modular forms and special cycles on Shimura curves*, Annals of Mathematics Studies, vol. 161, Princeton University Press, Princeton, NJ, 2006.
9. Sheng-Chi Liu, Riad Masri, and Matthew P. Young, *Subconvexity and equidistribution of Heegner points in the level aspect*, Compositio Mathematica **149** (2013), 1150–1174.
10. Philippe Michel, *The subconvexity problem for Rankin–Selberg L-functions and equidistribution of Heegner points*, Annals of Mathematics **160** (2004), no. 1, 185–236.
11. Oana Padurariu and Frederick Saia, *Bielliptic Shimura curves $X_0^D(N)$ with nontrivial level*, Research in Number Theory **11** (2025), no. 1, Paper No. 2, 23 pp., Correction, ibid. **11** (2025), no. 4, Paper No. 98; arXiv:2401.08829 [math.NT].
12. Hector Pasten, *Shimura curves and the abc conjecture*, Journal of Number Theory **254** (2024), 214–335.
13. Hector Pasten and Rocío Sepúlveda-Manzo, *On the abc and the abcd conjectures*, Bulletin of the Brazilian Mathematical Society, New Series **56** (2025), no. 4, 54, arXiv:2406.05083 [math.NT].
14. John G. Ratcliffe, *Foundations of hyperbolic manifolds*, third ed., Graduate Texts in Mathematics, vol. 149, Springer, Cham, 2019.
15. James Rickards, *Counting intersection numbers of closed geodesics on Shimura curves*, Research in Number Theory **9** (2023), no. 2, Article 20.
16. Andreas Wieser and Pengyu Yang, *A uniform Linnik basic lemma and entropy bounds*, 2026, Commentarii Mathematici Helvetici, published online January 22, 2026; arXiv:2201.05380 [math.NT].
17. Tonghai Yang, *An explicit formula for local densities of quadratic forms*, Journal of Number Theory **72** (1998), no. 2, 309–356.
18. Tonghai Yang, *Local densities of 2-adic quadratic forms*, Journal of Number Theory **108** (2004), no. 2, 287–345.
19. Xinyi Yuan and Shou-Wu Zhang, *On the averaged Colmez conjecture*, Annals of Mathematics **187** (2018), no. 2, 533–638.
20. Shou-Wu Zhang, *Heights of Heegner points on Shimura curves*, Annals of Mathematics **153** (2001), no. 1, 27–147.
21. Shou-Wu Zhang, *Equidistribution of CM-points on quaternion Shimura varieties*, International Mathematics Research Notices (2005), no. 59, 3657–3689.